Copenhagen Business School

Project

Data Management and Visualization

Optimizing Airbnb's Performance in Copenhagen: Data-Driven Insights for Growth & Success

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Executive Summary

Four years after the outbreak of the Covid-19 pandemic, tourism is recovering quickly, which brings the tourists and tourism revenue back to Copenhagen. Therefore, our project is to analyse and find critical success factors and growth opportunities for Airbnb in Copenhagen. The goal is to provide management-level executives, such as the Country Manager or Business Operation Lead, with a clear overview of the current state of business. Moreover, the pandemic is likely to cause significant changes in the tourism market, which leads the industry to adjust accordingly to maintain a strong player in the new competitive market. Accordingly, this study also aims to offer practical recommendations to the Operational and Business Development Teams to improve performance, increase revenue, and utilize new opportunities.

The project was conducted from a Data Scientist and Business Analyst perspective starting from the data preparation and cleaning. The data was cleaned to handle missing values, fix formatting issues, and remove outliers that could affect the results. Additional fields were created, such as average rating by neighbourhood and average rating per type of accommodation, to enhance the analysis. We subsequently implemented the exploratory analysis to discover some important findings. The data showed the mean price was 1360.41 DKK per night, with a standard deviation of 1478.57 DKK per night. The minimum price was 150 DKK, while the maximum price per night was 10,000 DKK. Furthermore, we can observe that a higher number of reviews also tends to have a higher rating, meaning that people have a good impression on those listings that are popular.

We eventually created three dashboards: the Overview Dashboard, Host Information Dashboard, and Neighbourhood Information Dashboard. These dashboards highlighted key market trends, such as the number of listings, average rating, and pricing across neighbourhood. They also analysed host performance metrics, including response times, superhost status, and guest reviews. Moreover, the project compared neighbourhoods based on availability, pricing, and guest ratings to identify top-performing areas and potential growth markets.

Based on our analysis, we recommend the following actions to help Airbnb improve its overall performance:

1. Assist hosts in enhancing their performance by reducing response times and encouraging them to achieve superhost status.
2. Focus on increasing experience ratings in the city center, which is Copenhagen's most profitable area, to ensure long-term business sustainability.
3. Promote affordable and high-potential neighborhoods, such as Valby, Vanlose, and Amager Ost, to attract budget-limited travelers.

These recommendations aim to enhance host performance and guest satisfaction while supporting Airbnb's sustainable growth and maximizing its potential in a competitive, data-driven market.

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1. Introduction

Airbnb was founded in 2008 by Brian Chesky, Nathan Blecharczyk and Joe Gebbia inspired by a situation when they hosted three guests on air mattresses in their San Francisco home, because they struggled to pay their rent (Airbnb, nd.). The name reflects the company's story, Airbnb is short for "airbed and breakfast" (IGMS, 2024). Since then it has revolutionized the short-term rental industry by connecting travellers and hosts from all over the world. It has more than 8 million active listings worldwide, with more than 5 million hosts and more than 2 billion guest arrivals all time (Airbnb, nd.). Airbnb's business can be defined as a multi-sided platform (Reinhold & Dolnicar, 2017), where costs and revenues can be found on both sides, as the platform has separate user groups on both sides. The users on both sides of the platform attract each other, which results in the "network effect" and makes the value of the platform highly dependent on the number of users (Eisenmann et al., 2006). The value increases as the platform satisfies the demand from both sides (Eisenmann et al., 2006). The company generates revenue by charging service fees to both parties, typically ranging from 3% to 14.2% for hosts and up to 20% for guests, depending on various factors (IGMS, 2024). As a result, it is critical to understand the key success factors and identify potential areas for development in order to maintain Airbnb's global growth and further enhance untapped opportunities.

As external Data Scientists, our role is to analyse and uncover critical success factors and growth opportunities within the Airbnb ecosystem. Our aim is to give executives at the management level, such as the Country Manager or Business Operation Lead a comprehensive picture of the present state of the business in the region and provide practical recommendations for the Operational Teams, as well as the Business Development Team to further enhance performance, maximize revenue and utilize untapped opportunities.

This project focuses on Copenhagen, a vibrant city renowned for its culture, sustainability, and innovation. As a prime example of a popular urban destination, Copenhagen's Airbnb market offers valuable insights into its potential. The analysis highlights market-specific trends and delivers actionable recommendations to shape Airbnb's strategies in similar urban markets worldwide. By doing so, it aims to strengthen Airbnb's position as a leader in the short-term rental industry. By analysing host

performance, neighbourhood trends, and overall market dynamics in Copenhagen, this study aims to draw insights that can inform both local and global strategies for improving performance and growing the platform.

The task was examined from a Data Scientist and Data Analyst position as well, by first preparing, and cleaning the dataset. The data was cleaned to address missing values, incorrect formatting, and outliers that could skew the results and several calculated fields were added, such as average ratings by neighbourhood and average rating per type of accommodation to enhance the analysis. We conducted an exploratory analysis to identify patterns, trends and correlations and created a clear, structured database to ensure that the analyses are consistent and reliable. Our goal was to transform the raw data into actionable insights and create dashboards that are intuitive and user-friendly, yet they still offer a comprehensive and complex picture of the current and potential trends of the market. We created three different dashboards, such as the Overview Dashboard, Host Information Dashboard and Neighbourhood Information Dashboard. We showcased high-level market trends, including the number of listings, average ratings, and pricing across neighbourhoods, focused on analysing host performance metrics such as response times, superhost status, and guest reviews and examined differences across neighbourhoods in terms of availability, pricing, and guest ratings to identify top-performing areas and potential markets for growth.

Airbnb hosts can leverage the insights to boost their performance and improve guest satisfaction. Operational teams can apply the findings to enhance the platform’s features and guide hosts toward adopting successful practices, while business development teams can utilise the data to strategically expand in Copenhagen and similar markets.

Overall, this project aims to deliver meaningful contributions and improve the performance of hosts and the experience of guests, while also driving Airbnb's sustainable growth and maximizing its potential as well as revenue in a competitive, data-driven market.

2. Data preparation & Exploratory Analysis

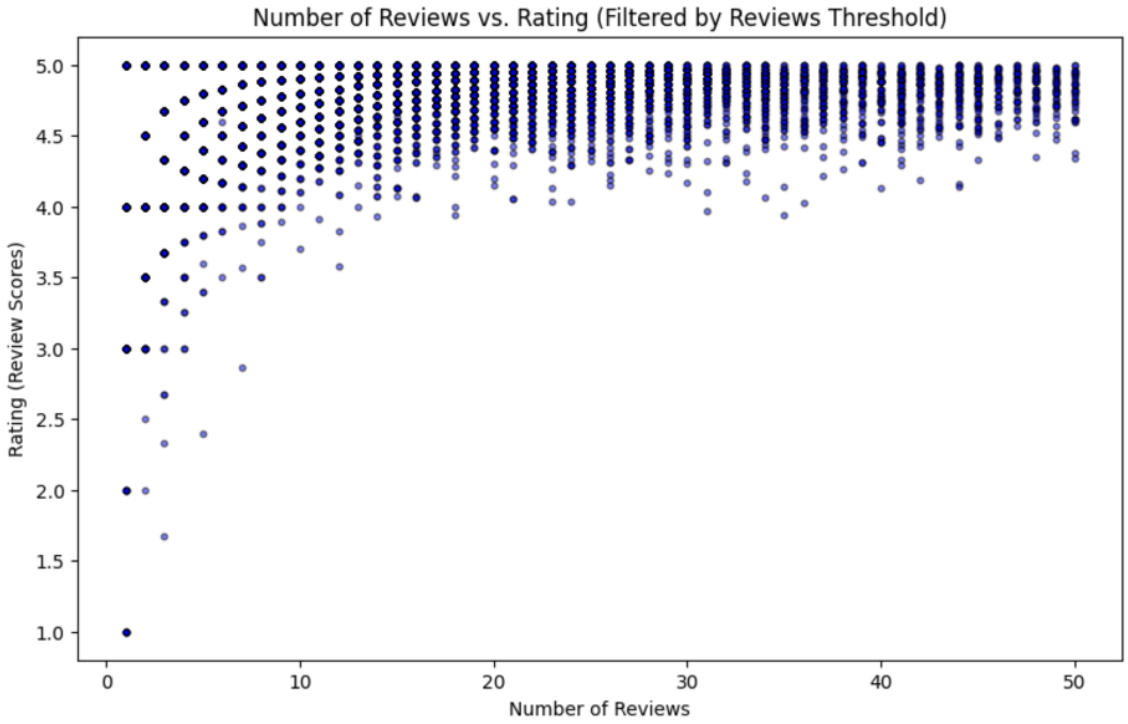
Our first goal was gaining a deeper understanding of the provided dataset through an exploratory analysis, as to determine which parameters are required to produce desired outcomes, and which were redundant. The dataset was to be cleaned, leaving only relevant entries. The process was to be done methodologically, following statistical guidelines. Furthermore, a dataset, which only includes relevant data increases efficiency in the later stages of our analysis, while at this stage, we were also able to pre-emptively expand the dataset with fields we were likely to utilise later on.

Upon first inspection, the dataset was rather vast, spanning 20909 rows and 75 columns. Immediately we were able to reduce the number of columns, as the columns calendar_updated, neighbourhood_group_cleansed, and license contained no data across any of the rows.

The dataset included no empty rows, however that does not indicate all of the 20909 carried statistically relevant or reliable information. With the relevant parameters for our analysis in sight, we conducted a basic statistical analysis of the remaining data, to determine the span of relevancy, focusing first on the number of reviews. Our hypothesis was that a substantial subset of our sample contained few reviews, thus reducing its relevance considering our goal. Indeed, while the mean number of reviews per accommodation across the dataset was 17.53 with a standard deviation of 49.69, 25% of the observed samples had less than 2 reviews, while 50% of all samples had less than 7 reviews.

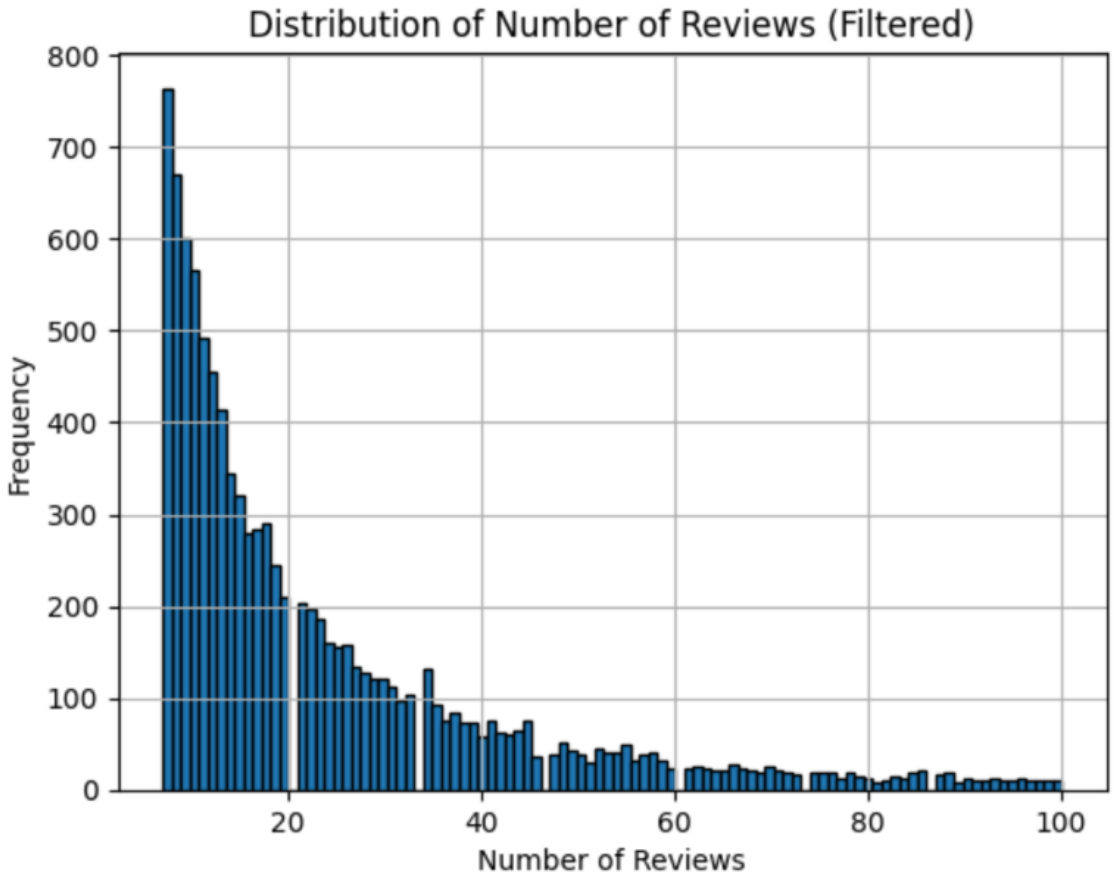
We then turned our attention to the descriptive statistics relating to the review_scores_rating column, which stores the information on the rating of the accommodations listed on Airbnb on a scale from 1 to 5. The mean rating was a surprisingly high 4.83 with a standard deviation of 0.26.

The following graph helps to illustrate the correlation of a samples' rating and the number of reviews it has received.



We can observe that the samples with a higher number of reviews also tend to have a higher rating, regressing towards the mean, calculated above.

Since the data on the number of reviews, a key metric for our analysis, is highly right skewed – the mean being significantly higher than the median, and the maximum number of 1972 reviews is much greater than the interquartile range, we decided to conduct further analysis on only the upper two quartiles – samples with 7 or more reviews. In this decision we considered the implications of small sample size bias – listings with so few reviews may not have enough data to accurately reflect guest satisfaction. Moreover, the above graph shows that listings with few reviews exhibit higher variability in rating, which might make them less reliable. By eliminating the long tail, we in turn eliminate potentially biased results. Splitting the data at the median – the 50% percentile, we may see more reliable patterns. This process is known in statistics as truncation, and is especially useful when dealing with skewed data, such as ours. The graph below shows exactly how skewed the distribution of number of reviews really is.



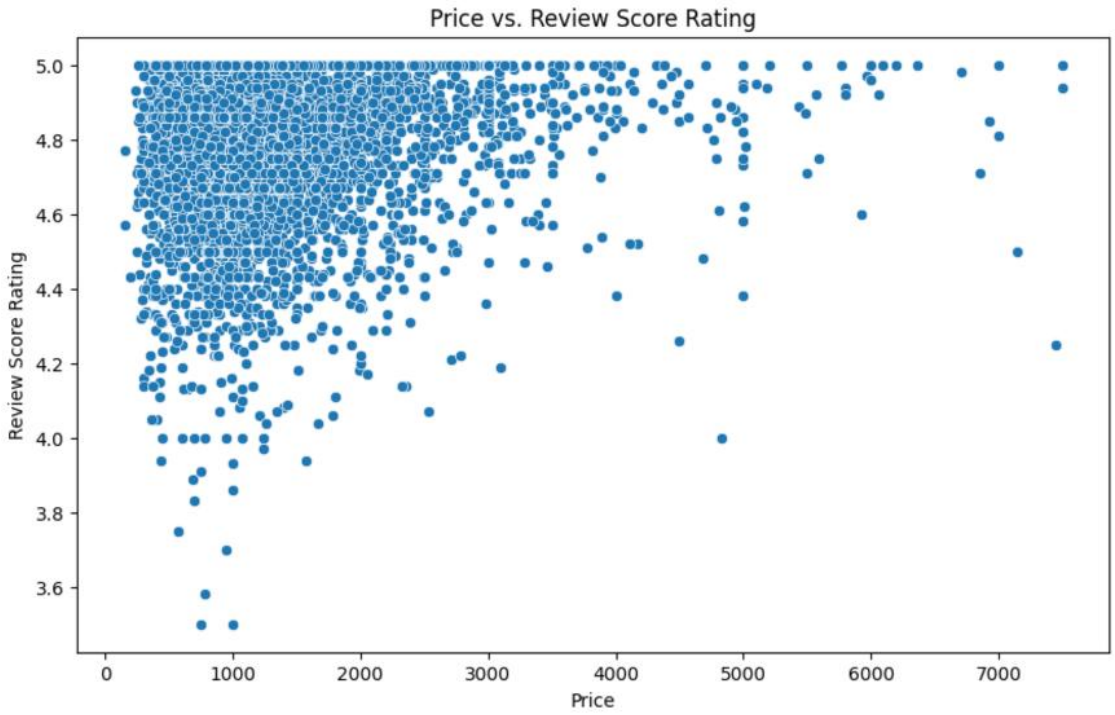
After eliminating all reviews with up to 7 reviews, we were left with 10503 rows and 72 columns. The impact of the decision delivered expected results. The mean number of reviews increased to 32.8 with a standard deviation of 54.7 and a median of 18 indicating a reduction in the skewness, furthermore, the long tail has been eliminated. We can determine, the data now focuses on more established patterns, we successfully reduced the impact of what we considered random noise.

The impact of our deletion of 50% of all entries had a slight but expected impact on review scores; the mean has decreased to 5.82 with a standard deviation of 0.17.

We now once again turned our attention to the 72 remaining columns and streamlined the dataset by eliminating those deemed irrelevant for establishing our goals, these include picture_url, host_picture_url, host_name, host_thumbnail_url, 'host_url, listing_url, neighbourhood, scrape_id. None of these columns provided any essential information which could be used in further analysis of visualisation of our results. The number of columns was reduced to 64.

Some columns, which carried essential information, however, contained some bothersome details, mainly the column ‘neighbourhood_cleansed’, which contained string type data on names of neighbourhoods where the listings were located, but the Danish letter Ø was left out of names like Nørrebro. We decided to replace the falsely spelled names with a standardised solution.

After having removed several columns, we pre-emptively added columns, which we deemed useful for further analysis. The column avg_rating_per_neighbourhood was created, where the average rating was displayed depending on the listed neighbourhood of each accommodation. The column avg_rating_per_room_type depicts the average rating per type of accommodation, which are discussed in previous chapters. These columns could serve for comparing a specific sample against the neighbourhood average, against the accommodation type average, or to compare neighbourhoods or accommodation types among each other.



We concluded our exploratory analysis with visualisation of correlation between a sample’s price and its rating. Plotting the two variables produces the following scatterplot. Calculating Pearson correlation coefficient, we come to a rather predictable value of 0.040 indicating a very weak, however positive correlation between the two variables at the p value of p = 0.00096, making the correlation statistically significant at 5% or 1%

significance level. We can thus conclude an increase in price does increase review scores very slightly, but we must reiterate this relationship is highly fragile despite statistical significance.

Simultaneously, the mean price of the 6716 samples which included a value in the column price, was 1360.41 Danish Kroner per night, with a standard deviation of 1478.57 dkk per night. The minimum was 150 dkk and can be observed on the above graph, while the maximum price per night was 10,000.00 Danish kroner per night.

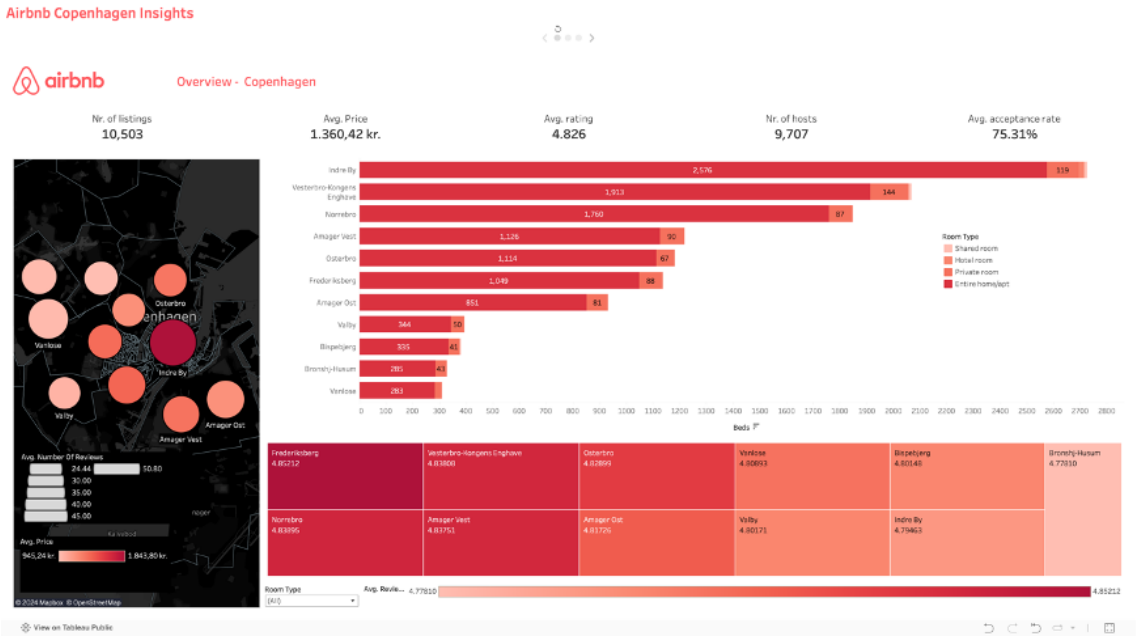
3. Database

To enable team members to access to the database and always have most up-to-date information after adjustments or updates having been made, a connected platform that the members can utilise is an ideal choice in terms of efficient teamwork. Therefore, the database that has undergone the preliminary cleaning process is imported into a PostgreSQL server, then the members can retrieve the data from one single source to avoid potential discrepancies in the data. Besides, the team members can do further cleaning work or add additional data if those actions are necessary. Since the data are stored in the PostgreSQL server, the data used for dashboards on Tableau is directly retrieved from the server by making connections between two ends. This is to ensure the accuracy of the data and to keep the data updated in real-time at all times.

4. Dashboards

4.1 Overview - Copenhagen Dashboard

The purpose of the Overview – Copenhagen dashboard is to provide an overall overview of Airbnb's listing information in Copenhagen. The dashboard is designed for executives at the management level, such as the Country Manager or Business Operation Lead. Accordingly, this dashboard serves as a strategic dashboard to monitor the long-term strategy for Airbnb and provide critical insights into the business. Moreover, the dashboard helps the company identify trends and make effective business decisions. The Overview – Copenhagen dashboard is shown in the following figure.



To show the characteristics of the Copenhagen market, five KPIs are displayed to the user, including the number of listings, average price, average rating, number of hosts, and average acceptance rate. These metrics are quantifiable measures of the availability and performance of listings and hosts. This information is shown at the top of the dashboard to give the user a quick and organized review. In this way, the massive data set turns into understandable measures that help executives identify key factors in the business.

Aside from the KPIs, the second visualization is the symbol map on the left side of the dashboard for showing quantitative data for individual locations. By hovering over different neighbourhoods, the user gets details on the average price and the average number of reviews. The map can allow the user to easily understand the information about an area. For example, the colour and the size represent two measures: average price and average number of reviews. If the neighbourhood is in a darker red, it has a higher average price. In a similar way, if the size of the neighbourhood circle is bigger, the average number of reviews of the neighbourhood is higher. As a result, we can gain some important business insights from the symbol map. For instance, the Indre By area, which is the city center, has the highest value on both measures. The information points out that visitors of Copenhagen would like to choose the city center as their accommodation location. The average price is relevant to Airbnb's revenue. Therefore, we can conclude that the Indre By is both popular and can bring a fortune to Airbnb. Moreover, the business operation team would like to find the area with potential. The map shows that

Vanlose and Amager Ost have a relatively high average number of reviews, however, both of them have low average prices. To be more specific, the company can promote these two areas to visitors who have a limited budget but still want to live in an attractive place. In conclusion, the symbol map is a great answer to the spatial question and can track areas that can give profit to the company.

To further understand the distribution of room types in different neighbourhoods in Copenhagen, a horizontal bar chart is displayed on the right side that shows the number of beds in different room types. The bar chart provides the company with a direct comparison of the quantity of beds and room types. From the bar chart, we can know that most of the listings in each neighbourhood are entire homes or apartments. Furthermore, the Indre By area offers the most beds, which suggests that the house owners in this area are more willing to rent out their houses. On the other hand, Vanlose has the least beds in Copenhagen, thus, the Business Development Team should focus on more development in this area to increase the number of hosts. To conclude, the bar chart of the beds displays the overview of the bed supply in Copenhagen for the managers to make a comparison and make an effective strategy based on the situation.

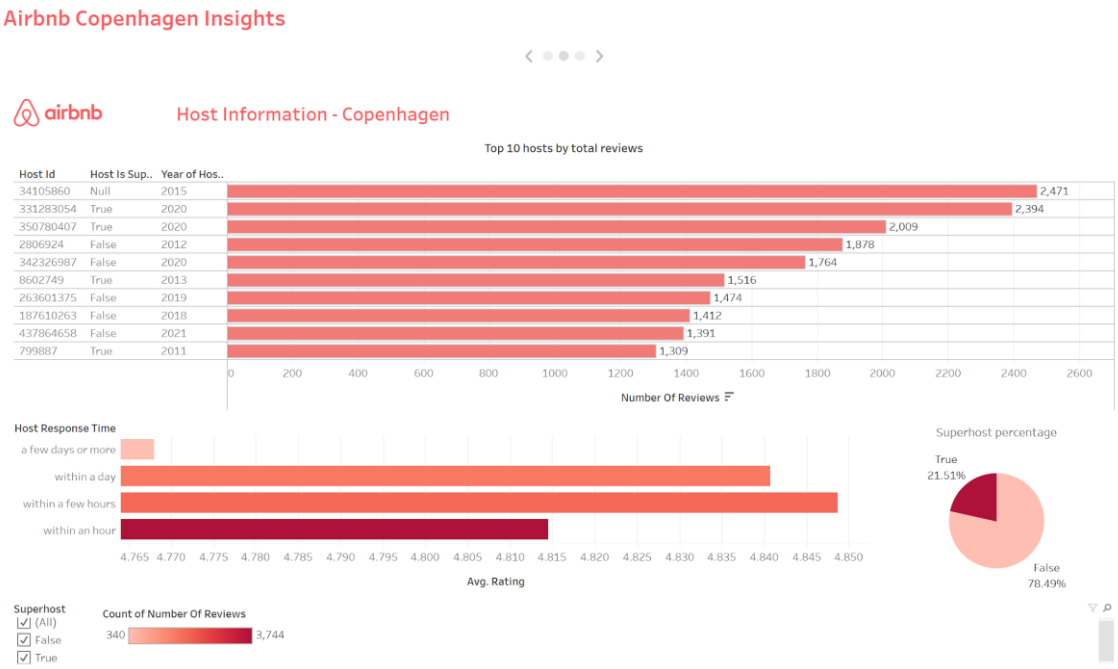
The last visualization under the bar chart is a tree map of each neighbourhood in Copenhagen with respect to the average rating score. The colour and the size of each rectangle in the tree map can represent the degree of average rating score. The size of different neighbourhoods is quite similar, which points out that the average rating is around 4.8 in all areas. However, the colour shows more highlights for the user to know the differences of the average rating between neighbourhoods. To be more specific, we can know that the Frederiksberg area has the highest rating, in contrast, the Bronshoj-Husum is the lowest-rating neighbourhood in Copenhagen. Therefore, the business operation team can make a strategy to maintain the high-rating area and improve the low-rating neighbourhood. Moreover, an interactive element is provided by the filter room type, which allows the user to change between different room types and understand the average rating under the filtered situation. As a result, it is easy for the managers to observe the detailed information of average ratings by neighbourhoods with the tree map.

In conclusion, the Overview – Copenhagen dashboard is designed to give an introduction to the people at the management level to understand the current situation of Airbnb's

operation in Copenhagen. All sheets in the dashboard provide key information that allows Airbnb to look closely into their business in a more organized way and make some data-driven strategies.

4.2. Host Information Dashboard

The aim of the Host Information Dashboard is to explore the key attributes and characteristics of Airbnb hosts in Copenhagen and identify the potential success factors in order to optimize performance, enhance guest satisfaction and provide actionable insights for achieving higher engagement and competitive advantage in the marketplace. It focuses on metrics such as ratings, response time and superhost status. This dashboard provides useful insights into the performance of hosts and engagement of guests with answering questions such as: what causes hosts to have high ratings? How does response time affect guest satisfaction ratings? And what impact does achieving superhost status have on host performance? This dashboard is designed for Airbnb's Operational Teams and the Business Development Team, and provides actionable insights into areas where hosts can be supported to improve performance, and highlights the value of promoting superhost status. It provides a practical tool for hosts to benchmark their performance against their top-rated peers and understand how factors such as response time and superhost recognition can increase guest satisfaction. The dashboard can be seen on the following figure.



It is built around several key components and one of its main features is the "Top 10 hosts by Total Reviews" bar chart, which highlights the most active and highly rated hosts in the market. The hosts are ranked based on the number of reviews they received, which can be a good indicator of the number of bookings as well, since a guest can only rate the listing if they completed a stay there. The top hosts received between 1309 and 2471 reviews, which indicates their dominant presence in the market. This visualisation highlights success factors and provides benchmarks for other hosts who want to increase guest engagement. Four of the top 10 hosts are super hosts, including the second and third in the ranking. It can be observed that they have been using Airbnb since 2020, while the first host has been present since 2015, yet they do not have a comparative advantage. This could indicate that being a superhost attracts more guests.

Another component of the dashboard is focused on the response time of hosts and links it with the average ratings of guests as well as the number of reviews. It is represented by a horizontal bar chart and the colour of the bars indicate the number of reviews. By hovering over the bars, a tooltip is displayed summarizing the response time, number of reviews and average ratings. The data show a clear trend, since it is easily observed that faster response times lead to higher ratings. For example, hosts who respond "within an hour" achieve the highest average rating of 4.81. This highlights the importance of fast communication if we want to improve overall ratings and guest satisfaction.

In addition, a pie chart represents the distribution of hosts and superhosts. We can see that only 21,51% of hosts have superhost status, which emphasizes the relatively exclusive nature of this distinction. Users can also use an interactive filter that allows them to filter their results based on superhost status. This allows them to take a closer look at the performance differences between superhosts and non-superhosts and highlight opportunities for others to follow well-performing success factors. The charts show that superhosts tend to respond much faster and have relatively higher average ratings.

This dashboard was designed to facilitate easy understanding and straightforward insights, while representing key insights. Each visualization contributes to the narrative as the bar charts show clear trends in review numbers, ratings and response times, and the pie chart summarizes the distribution of superhosts and non-superhosts. It is easily observed that the success of hosts is highly dependent on the response rate and superhost

status. Hosts who communicate quickly, especially those who respond within an hour, achieve significantly higher ratings, while super-hosts benefit from greater visibility and engagement. These charts illustrate the advantages of the superhost status and the numerical dominance of non-hosts only indicate the significant potential for improving their performance by achieving the superhost recognition. The non-superhosts account for the majority of hosts, yet they often fall behind in performance indicators.

We can conclude that hosts need to focus on their response times and aim to respond to guest inquiries within an hour to maximize guest satisfaction, because it is correlated with higher average ratings. To achieve this, automated messaging tools could be implemented in order to ensure timely replies, even when unavailable. Hosts should be also encouraged to strive for a superhost status as it enhances the number of bookings and satisfaction as well. In order to achieve this they need to focus on maintaining a 4.8 or higher overall rating, a 90% or higher response rate, a less than 1% cancellation rate and have at least 10 reservations, or 3 reservations that total at least 100 nights (Airbnb, n.d.). By regularly monitoring Airbnb's performance indicators and ensuring best practices, these conditions should not be a challenge to achieve, as the average overall rating is already 4.826.

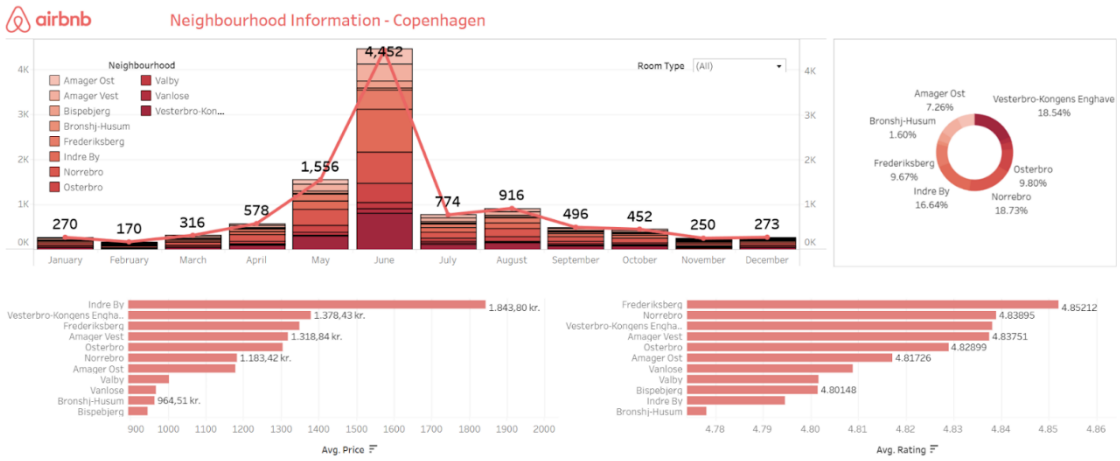
To enhance performance and increase revenue, Airbnb should consider actively promoting the benefits of superhost status. Educating non-superhosts on its advantages, such as increased satisfaction, greater visibility and higher number of bookings, could encourage them to strive for this recognition. Effective strategies could include hosting webinars, sending performance reports and providing tips on how to achieve the superhost status, as well as offering tutorials or online training for new and non-superhost hosts on how to improve communication, manage guest experiences, and optimize listings. Airbnb could also integrate user-friendly tools to visually display KPIs such as average response rate, average ratings and guest reviews. Recognizing and rewarding superhosts with additional incentives, such as reduced fees, exclusive promotions or targeted ads for their listings, could further motivate hosts to improve their performance. To support faster communication, Airbnb could develop features like customizable response templates, automatic replies, or reminders for the hosts in case of delayed responses.

All in all, the Host Information Dashboard provides valuable insights for a wide range of stakeholders. For Airbnb's operational teams, the dashboard highlights the importance of recognizing and promoting superhosts, while also identifying areas where the broader host base can be improved. It highlights key success factors that are essential for identifying areas for growth and business development in key markets such as Copenhagen. By focusing on key areas such as reviews, response times and the impact of superhosts, it not only sheds light on current trends, but also highlights possible strategic improvements that will ultimately drive the success of both hosts and the Airbnb platform. Airbnb needs to promote the superhost status, including the rules for achieving it and hosts need to be encouraged to improve their overall performance and aim for a superhost status.

4.3. Neighbourhood Information Dashboard

The third dashboard is mainly aimed at the internal Business Development team to grasp the main respects of listings of the ten administrative districts in Copenhagen, and then to discover potential business opportunities to better drive up the overall leasing market size. This dashboard can serve as an analytical dashboard as it allows team members to delve into Airbnb's business performance over the years in both Copenhagen and Frederiksberg municipalities, the latter of which is usually considered a part of the former by the public.

It is sometimes the case that individuals, who are not from the business development background, would also like to know fundamental information about the neighbourhoods in Copenhagen; therefore, the dashboard is divided into four sections with appropriate charts, and each section focuses on one main metric, making it much easier for members from different teams to easily understand the data presented. The *Neighbourhood Information – Copenhagen* dashboard is shown in the following figure.



The dashboard *Neighbourhood Information – Copenhagen* contains information on the number of accommodation bookings by month, the proportion of the listings posted in each area, the average of accommodation prices, and the average of house/room reviews in the neighbourhood. Firstly, in the top-left section we can see a bar chart showing the percentage of bookings from different administrative districts in each month, with a line chart presenting the trend of booking popularity in Copenhagen. In this section, there is a 'Room Type' filter, which allows team members to view the listing conditions by choosing their preferred type, and then further analyse the figures in the chart. Secondly, a pie chart lies to the right of the first chart mentioned. It describes the total number of listings available by neighbourhood as percentage, which the team members can easily tell which areas need some boosters, and the areas are relatively mature from.

By default, we can see from the first chart that the booking number starts to noticeably increase in May and then surge to the highest point during the year, rising from 1,556 to 4,452. After that, the figure drastically goes down to less than a fifth of the level from the previous month, before slightly growing back to approximately 900 in August. In the course of the months from May to August, it is obvious the tourist season for Copenhagen. Looking at the chart for June, we can discover that most travellers choose to stay in the city centre (Indre By), Vesterbro/Kongens Enghave, and Nørrebro, which partially match the pattern of the pie chart to some extent. Besides, it can also be discerned that the majority of room types in Copenhagen are comprised of 'Entire home/apartment' and 'Private room.' However, the room types 'Hotel room' and 'Shared room' are relatively few, implying that most travellers visiting Copenhagen prefer private space during their

stay. These are explicit information that individuals can understand from the upper section of the dashboard.

In terms of business development, there is a rapidly increasing number of tourists coming to Denmark for summer holidays because of mild temperatures and extended daylight hours (Roselyne Min, 2024), the business development are encouraged to approach individuals who own spare properties and convince them to put their houses into the market. In particular, those who have both summer houses in the countryside and properties in Copenhagen are greatly potential target landlords because it is highly likely that they will not be staying in the city during the summer holiday.

In the lower section of the dashboard, there are two bar charts showing the average of accommodation prices and the average of the reviews that guests leave according to the experience during their stay in each neighbourhood. The 'Avg. Price' chart is located in the bottom-left section. The team members can understand the price range of the listings by looking at the chart; then, use this information to analyse the market condition and develop customised strategies targeted at different areas.

Next to the diagram, there is another bar chart comparing the average of ratings in the neighbourhood. As most travellers predominantly depend on the ratings when deciding where they are going to stay during their visit, this is a crucial figure not only for the tourists but also for internal team members, especially the business development team in terms of business operation in general. With these two bar charts, the team member can segment the Copenhagen market into certain categories and make plans to further improve the quality of the listings, or provide suggestions for hosts in order to optimise the experience of the future guests.

By comparing the two charts, it can be seen that the average price in the city centre is the highest, but the rating for the city centre is the bottom two. We can presume that people choosing to stay in the city center expect a high-quality living environment given the fact they are willing to rent expensive properties in the city. Easy access to the most tourist attractions and public transport, which supports the high prices, is the benefit of staying in the city centre. However, landlords are also advised to improve the quality of their houses so as to raise the overall experience ratings for sustainable business in the long-term. Additionally, properties in areas, such as Valby or Vanlose, are much more

affordable compared to those in the surrounding of the city centre. Since the average rating figures in the different districts are quite similar, staying outside the city centre is a budget-friendly option for people who would not like to spend much on accommodation. This also might advance the suburbs and expand the entire leasing market.

On the whole, this dashboard depicts general information that is important for the business development team to make strategic decisions, and also can be presented to other teams for the fundamental understanding of the leasing market in Copenhagen.

5. Conclusion

This project has embodied the essence of data science and business analytics, ranging from data preparation and data cleaning to data transformation and exploratory analysis. The end results are three straightforward dashboards with material information that is demonstrated in the form of charts or diagrams with metrics, which make it more effortless for the target audience of the project, such as the management, members of the operation team and business development team, to obtain the overview of Airbnb's accommodation condition in Copenhagen and understand the insights of the data presented on the dashboards. This analytical tool facilitates the growth and improvement of the business in the city as it provides staff members with access to important quantitative measures, including the average price, the average ratings, the host response rates and the number of listings in the neighbourhoods. With these figures, the teams can make better decisions that are suitable for different areas/situations.

Overall, those hosts who have not meet the requirements of super hosts should be encouraged to reply to guest messages within an hour, provide pleasant living environment and sufficient equipment to the extent that the accommodation costs of staying their properties are considered reasonable, leading to better rating scores which are the most important factor in the concept of Airbnb. Furthermore, internal staff should explore more business opportunities in the districts, such as Valby, Vanlose, and Amager Ost, to scale up the operations and the market size in Copenhagen. Lastly, since the business core is the city centre area, it is also critical to keep the current performance of the area and to provide suggestions for the hosts to tailor the listing prices to the seasonal

needs and newly formed travel trends, resulting in much greater profits that support the development and advancement of Airbnb's operations in Copenhagen.

Appendix

Field	Type	Description
id	integer	Airbnb's unique identifier for the listing
last_scraped	datetime	UTC. The date and time this listing was "scraped".
source	text	How was the data acquired - scraped
name	text	Name of the listing
description	text	Detailed description of the listing
neighborhood_overview	text	Host's description of the neighbourhood
host_id	integer	Airbnb's unique identifier for the host/user
host_since	date	The date the host/user was created.
host_location	text	The host's self-reported location
host_about	text	Description about the host
host_response_time	numeric	The time in which the host responds
host_response_rate	text	The rate at which a host responds to booking requests.
host_acceptance_rate	numeric	The rate at which a host accepts booking requests.
host_is_superhost	boolean	[t=true; f=false]
host_neighbourhood	text	Host's neighbourhood
host_listings_count	text	The number of listings the host has (per Airbnb calculations)
host_total_listings_count	text	The number of listings the host has (per Airbnb calculations)
host_verifications	text	
host_has_profile_pic	boolean	[t=true; f=false]
host_identity_verified	boolean	[t=true; f=false]
neighbourhood_cleansed	text	The neighbourhood as geocoded using latitude and longitude
latitude	numeric	Uses WGS84 projection for latitude
longitude	numeric	Uses WGS84 projection for longitude
property_type	text	Self-selected property type
room_type	text	Entire home/apt|Private room|Shared room|Hotel
accommodates	integer	The maximum capacity of the listing
bathrooms	numeric	The number of bathrooms in the listing
bathrooms_text	string	Textual description of the number of bathrooms

bedrooms	integer	The number of bedrooms
beds	integer	The number of beds
amenities	json	
price	currency	Daily price in local currency
minimum_nights	integer	Minimum number of night stay
maximum_nights	integer	Maximum number of night stay
minimum_minimum_nights	integer	The smallest minimum_night value from the calendar (looking 365 nights in the future)
maximum_minimum_nights	integer	The largest minimum_night value from the calendar (looking 365 nights in the future)
minimum_maximum_nights	integer	The smallest maximum_night value from the calendar (looking 365 nights in the future)
maximum_maximum_nights	integer	The largest maximum_night value from the calendar (looking 365 nights in the future)
minimum_nights_avg_ntm	numeric	The average minimum_night value from the calendar (looking 365 nights in the future)
maximum_nights_avg_ntm	numeric	The average maximum_night value from the calendar (looking 365 nights in the future)
has_availability	boolean	[t=true; f=false]
availability_30	integer	Availability in the next 30 days
availability_60	integer	Availability in the next 60 days
availability_90	integer	Availability in the next 90 days
availability_365	integer	Availability in the next 365 days
calendar_last_scraped	date	The date of last data scrape
number_of_reviews	integer	Number of reviews
number_of_reviews_ltm	integer	The number of reviews the listing has (in the last 12 months)
number_of_reviews_l30d	integer	The number of reviews the listing has (in the last 30 days)
first_review	date	Date of the first review
last_review	date	Date of the last review
review_scores_rating	numeric	Rating based on several parameters
review_scores_accuracy	numeric	Rating of description accuracy
review_scores_cleanliness	numeric	Rating of cleanliness
review_scores_checkin	numeric	Rating of check in process
review_scores_communication	numeric	Rating of host's communication

review_scores_location	numeric	Rating of accomodation location
review_scores_value	numeric	Rating of accomodation value
instant_bookable	boolean	[t=true; f=false]. Whether booking is automatic without host approval
calculated_host_listings_count	Numeric	Number of listings by the host
calculated_host_listings_count_entire_homes	Numeric	Number of entire home listings by the host
calculated_host_listings_count_private_rooms	Numeric	Number of private room listings by the host
calculated_host_listings_count_shared_rooms	Numeric	Number of shared room listings by the host
reviews_per_month	numeric	Number of reviews over listing lifetime
avg_rating_per_neighbourhood	numeric	Average rating per neighbourhood
avg_rating_per_room_type	numeric	Average rating per room type

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